

Ayush Parashar

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Education

SGT University

June 2027

B.Tech in Computer Science with specialization in AI and ML

Gurugram, Haryana

- **Relevant Coursework:** Data Structures and Algorithms (C++), Prob & Stat in CS (Python), Intro to CS II (C++), Linear Algebra w/Computational Applications (Python)

Experience

Samatrix Consulting pvt ltd

June 2025 – July 2025

Data Analyst and Statistical Modeling intern

Gurugram, Haryana

- Optimized call center staffing using M/M/s queue simulation, reducing simulated 95th percentile customer wait time by 40% by identifying the optimal agent count (4 agents vs 1).
- Designed and analyzed A/B testing framework (10,000+ users per group) using Z-test and confidence intervals, identifying a statistically significant 2% uplift in conversion rate ($p < 0.001$), guiding data-driven product decisions.
- Built survival analysis models using Kaplan–Meier and Cox Proportional Hazards, identifying treatment groups with significantly higher median survival times and hazard reduction, enabling evidence-based healthcare insights.
- Implemented Statistical Process Control using P-charts and Western Electric rules, detecting out-of-control defect spikes and simulated process shifts, improving anomaly detection sensitivity by 30%.
- Developed portfolio risk modeling framework calculating 95% 1-day Value at Risk (VaR) using Normal, Student's t-distribution, and Historical Simulation, improving extreme-risk estimation accuracy through heavy-tail modeling.

Projects

KindMinds AI | Next.js, Supabase, Python, AI

- Engineered a stateful conversational AI application with persistent memory for context-aware multi-turn conversations.
- Implemented modular dialogue architecture with controlled response generation and Supabase-powered session persistence.
- Developed a Python AI backend with real-time synchronization and integrated it with a responsive Next.js frontend.

Movie Recommender System | Python, Pandas, Scikit-learn, NLTK

- Built a content-based recommendation system for 4,800+ TMDB movies using genre, cast, keywords, and director metadata.
- Implemented NLTK preprocessing, CountVectorizer, and cosine similarity, reducing recommendation retrieval time by 35%.

Deep Learning Fashion Recommender System | Python, TensorFlow, Keras, ResNet50, Streamlit

- Built an image-based recommendation engine using pretrained ResNet50 to generate 2048-dimensional product embeddings.
- Implemented L2 normalization and KNN similarity search for real-time top-5 visual recommendations across 5,000+ images.
- Precomputed embeddings, reducing inference latency by 60%, and deployed an interactive Streamlit interface.

Technical Skills

Languages: Python, C++, Kotlin, R, Dart, Java

Technologies: Django, Flask, TensorFlow, PyTorch, NumPy, Pandas, Matplotlib, Android SDK, Flutter

Concepts: Compiler, Operating System, Virtual Memory, Cache Memory, Encryption, Decryption, Artificial Intelligence, Machine Learning, Neural Networks, API, Database Normalization, Agile Methodology, Cloud Computing